# Srednicki QFT: Chapter 38

Spinor Technology

Cadabra2 expressions and numerical verification

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# Contents

# 1 $\sigma^\mu$ Completeness Relation

## 1.1 Main identity

The $\sigma$ matrices form a complete basis for $2 \times 2$ complex matrices. The completeness relation is (Srednicki eq. 38.1):

$$\boxed{\sigma^\mu{}_{\alpha\dot\alpha}\, \sigma_\mu{}^{\beta\dot\beta} = -2\, \delta_\alpha{}^\beta\, \delta_{\dot\alpha}{}^{\dot\beta}} \tag{38.1}$$

where $\sigma_\mu{}^{\beta\dot\beta} = g_{\mu\nu}\bar\sigma^{\nu\dot\beta\beta}$ (both spinor indices raised by $\varepsilon$).

✓ Max error: $0.0e + 00$.

The nonzero components of the LHS are: $(\alpha, \dot\alpha, \beta, \dot\beta) \in \{(1, 1, 1, 1), (1, 2, 1, 2), (2, 1, 2, 1), (2, 2, 2, 2)\}$, each equal to $-2$.

## 1.2 Alternative form

Equivalently (eq. 35.5):

$$\varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\, \sigma^\mu{}_{\alpha\dot\alpha}\, \sigma^\nu{}_{\beta\dot\beta} = -2\, g^{\mu\nu} \tag{35.5}$$

**Cadabra2:** $\epsilon^{\alpha\beta}\epsilon^{\dot\alpha\dot\beta}\sigma^{\mu}{}_{\beta\dot\beta} = \bar\sigma^{\mu\dot\alpha\alpha}$

✓ Max error vs $-2g^{\mu\nu}$: $0.0e + 00$.

# 2 Trace Identities

## 2.1 Two-sigma trace

The fundamental trace identity (Srednicki eq. 38.5):

$$\boxed{\mathrm{Tr}[\sigma^\mu\bar\sigma^\nu] \equiv \sigma^\mu{}_{\alpha\dot\alpha}\, \bar\sigma^{\nu\dot\alpha\alpha} = -2\, g^{\mu\nu}} \tag{38.5}$$

**Cadabra2 summand:** $\sigma^{\mu}{}_{\alpha\dot\alpha}\bar\sigma^{\nu\dot\alpha\alpha}$

Explicit matrix (rows $\mu$, columns $\nu$):

$$\mathrm{Tr}[\sigma^\mu\bar\sigma^\nu] = \begin{pmatrix} +2.0 & +0.0 & +0.0 & +0.0 \\ +0.0 & -2.0 & +0.0 & +0.0 \\ +0.0 & +0.0 & -2.0 & +0.0 \\ +0.0 & +0.0 & +0.0 & -2.0 \end{pmatrix} = -2g^{\mu\nu}$$

✓ $\mathrm{Tr}[\sigma^\mu\bar\sigma^\nu] = -2g^{\mu\nu}$: max error $0.0e + 00$.

✓ $\mathrm{Tr}[\bar\sigma^\mu\sigma^\nu] = -2g^{\mu\nu}$: max error $0.0e + 00$.

**Physical interpretation:** $\sigma^\mu$ and $\bar\sigma^\nu$ are "dual" bases for $2 \times 2$ matrices; the trace gives their inner product $-2g^{\mu\nu}$. Any $2 \times 2$ matrix $A$ expands as $A = -\frac{1}{2} \sum_\mu \bar\sigma^\mu\, \mathrm{Tr}[\sigma^\mu A]$.

### 2.2 Four-sigma trace

The four-sigma trace identity (eq. 38.6), with Srednicki's mostly-plus metric $g = \mathrm{diag}(-1, +1, +1, +1)$:

$$\boxed{\mathrm{Tr}[\sigma^\mu \bar\sigma^\nu \sigma^\rho \bar\sigma^\sigma] = 2\big(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}\big) + 2i\,\varepsilon^{\mu\nu\rho\sigma}} \tag{38.6}$$

$$\mathrm{Tr}[\bar\sigma^\mu \sigma^\nu \bar\sigma^\rho \sigma^\sigma] = 2\big(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}\big) - 2i\,\varepsilon^{\mu\nu\rho\sigma} \tag{38.6$'$}$$

(The conjugate trace carries $-2i$ instead of $+2i$; the sign difference is specific to the mostly-plus convention.)

**Sample numerical values** ($\varepsilon^{0123} = +1$):

| $(\mu, \nu, \rho, \sigma)$ | Computed | Expected | |
|---|---|---|---|
| $(0, 1, 2, 3)$ | $+0.00 + +2.00i$ | $+0.00 + +2.00i$ | ✓ |
| $(1, 0, 2, 3)$ | $+0.00 + -2.00i$ | $+0.00 + -2.00i$ | ✓ |
| $(0, 1, 3, 2)$ | $+0.00 + -2.00i$ | $+0.00 + -2.00i$ | ✓ |
| $(0, 2, 1, 3)$ | $+0.00 + -2.00i$ | $+0.00 + -2.00i$ | ✓ |
| $(1, 2, 3, 0)$ | $+0.00 + -2.00i$ | $+0.00 + -2.00i$ | ✓ |
| $(2, 3, 0, 1)$ | $+0.00 + +2.00i$ | $+0.00 + +2.00i$ | ✓ |

✓ All 256 components verified. Max error: $0.0e + 00$. ✓ Conjugate trace: max error $0.0e + 00$.

## 3 Van der Waerden Index Gymnastics

### 3.1 Raising and lowering spinor indices

The $\varepsilon$-metric raises and lowers spinor indices (Srednicki eqs. 38.7–38.13):

$$\psi_\alpha = \varepsilon_{\alpha\beta}\,\psi^\beta \qquad \text{(lower undotted)}, \tag{38.7}$$
$$\psi^\alpha = \varepsilon^{\alpha\beta}\,\psi_\beta \qquad \text{(raise undotted)}, \tag{38.8}$$
$$\chi_{\dot\alpha} = \varepsilon_{\dot\alpha\dot\beta}\,\chi^{\dot\beta} \qquad \text{(lower dotted)}, \tag{38.9}$$
$$\chi^{\dot\alpha} = \varepsilon^{\dot\alpha\dot\beta}\,\chi_{\dot\beta} \qquad \text{(raise dotted)}, \tag{38.10}$$

with the Srednicki convention $\varepsilon_{12} = -1,\ \varepsilon^{12} = +1$ ($\varepsilon_{21} = +1,\ \varepsilon^{21} = -1$).

Key consistency: $\varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta^\alpha_\gamma$. ✓ $\varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta^\alpha_\gamma$: max error $0.0e + 00$.

### 3.2 Raising both spinor indices of $\sigma^\mu$

The relation between $\sigma$ and $\bar\sigma$ via index raising (eq. 38.13):

$$\bar\sigma^{\mu\dot\alpha\alpha} = \varepsilon^{\alpha\beta}\,\varepsilon^{\dot\alpha\dot\beta}\,\sigma^\mu{}_{\beta\dot\beta} \tag{38.13}$$

**Cadabra2:** $\epsilon^{\alpha\beta}\epsilon^{\dot\alpha\dot\beta}\sigma^\mu{}_{\beta\dot\beta} = \bar\sigma^{\mu\dot\alpha\alpha}$

✓ All four $\mu$ values verified: max error $0.0e + 00$.

# 4 Spinor Bilinears

## 4.1 Lorentz transformation properties

The key bilinear forms and their Lorentz transformation properties:

$$\chi\psi \equiv \chi^\alpha \psi_\alpha = \varepsilon^{\alpha\beta}\chi_\alpha\psi_\beta \qquad \text{Lorentz scalar (left-handed)} \tag{38.14a}$$

$$\chi^\dagger\psi^\dagger \equiv \chi^\dagger_{\dot\alpha}\psi^{\dagger\dot\alpha} \qquad \text{Lorentz scalar (right-handed)} \tag{38.14b}$$

$$\chi^\dagger_{\dot\alpha}\,\bar\sigma^{\mu\dot\alpha\alpha}\,\psi_\alpha \qquad \text{Lorentz 4-vector} \tag{38.14c}$$

**Cadabra2 vector bilinear:** ${}_{\dot\alpha}\bar\sigma^{\mu\dot\alpha\alpha}\psi_\alpha$

## 4.2 Momentum matrix in spinor space

The momentum $p^\mu$ maps to a $2\times 2$ spinor matrix:

$$p_{\alpha\dot\alpha} = p_\mu\,\sigma^\mu{}_{\alpha\dot\alpha} = -E\,I_2 + \mathbf{p}\cdot\boldsymbol{\sigma}, \tag{1}$$

with $\det(p_{\alpha\dot\alpha}) = -p^2/4$ (up to normalisation). For massless particles: det$=0$, so $p_{\alpha\dot\alpha}$ factorises as $p_{\alpha\dot\alpha} = \lambda_\alpha\tilde\lambda_{\dot\alpha}$ — the starting point for spinor-helicity methods.

## 4.3 Gordon identity

In 4-component Dirac notation, the vector bilinear decomposes as:

$$\bar u(p')\,\gamma^\mu\,u(p) = \bar u(p')\left[\frac{p'^\mu + p^\mu}{2m} + \frac{i\sigma^{\mu\nu}(p'-p)_\nu}{2m}\right]u(p) \tag{2}$$

In 2-component language, the vector part comes from $\tilde u^{\dot\alpha s}\bar\sigma^{\mu\dot\alpha\alpha}u^s_\alpha$ and the tensor part from the generators $S_L^{\mu\nu}$.

# 5 Fierz Identities

## 5.1 Schouten identity

The $\varepsilon$-Fierz or Schouten identity:

$$\boxed{\varepsilon_{\alpha\gamma}\,\varepsilon_{\beta\delta} = \varepsilon_{\alpha\beta}\,\varepsilon_{\gamma\delta} - \varepsilon_{\alpha\delta}\,\varepsilon_{\gamma\beta}} \tag{38.17}$$

**Cadabra2:** $\epsilon_{\alpha\gamma}\epsilon_{\beta\delta} = \epsilon_{\alpha\beta}\epsilon_{\gamma\delta} - \epsilon_{\alpha\delta}\epsilon_{\gamma\beta}$

✓ Schouten identity: max error $0.0e+00$.

## 5.2 $\sigma$-matrix Fierz

From the completeness relation (§1), the $\sigma$-matrix Fierz identity is:

$$\left(\bar\sigma^\mu\right)^{\dot\alpha\alpha}\left(\sigma_\mu\right)_{\beta\dot\beta} = -2\,\delta^\alpha_\beta\,\delta^{\dot\alpha}_{\dot\beta} \tag{38.18}$$

### 5.3 Four-Fermi Fierz

For four Grassmann-valued spinors:

$$\left(\chi^\dagger_{\dot\alpha}\bar\sigma^{\mu\dot\alpha\alpha}\psi_\alpha\right)\left(\xi^\dagger_{\dot\beta}\bar\sigma_\mu{}^{\dot\beta\beta}\eta_\beta\right) = -2\left(\chi^\dagger_{\dot\alpha}\xi^{\dagger\dot\alpha}\right)\left(\eta^\beta\psi_\beta\right) \tag{38.21}$$

This follows from the $\sigma$-Fierz plus Grassmann anticommutativity, and is used to rewrite four-fermion interaction terms.

## 6 Numerical Verification Summary

| Identity | Equation | Max error |
|---|---|---|
| $\sigma^\mu{}_{\alpha\dot\alpha}\sigma_\mu{}^{\beta\dot\beta} = -2\delta^\beta_\alpha\delta^{\dot\beta}_{\dot\alpha}$ | (38.1) | $0.0e+00$ |
| $\varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\sigma^\mu_{\alpha\dot\alpha}\sigma^\nu_{\beta\dot\beta} = -2g^{\mu\nu}$ | (35.5) | $0.0e+00$ |
| $\mathrm{Tr}[\sigma^\mu\bar\sigma^\nu] = -2g^{\mu\nu}$ | (38.5) | $0.0e+00$ |
| $\mathrm{Tr}[\bar\sigma^\mu\sigma^\nu] = -2g^{\mu\nu}$ | (38.5) | $0.0e+00$ |
| $\mathrm{Tr}[\sigma^\mu\bar\sigma^\nu\sigma^\rho\bar\sigma^\sigma] = 2(\cdots) + 2i\varepsilon$ | (38.6) | $0.0e+00$ |
| $\mathrm{Tr}[\bar\sigma^\mu\sigma^\nu\bar\sigma^\rho\sigma^\sigma] = 2(\cdots) - 2i\varepsilon$ | (38.6′) | $0.0e+00$ |
| $\varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta^\alpha_\gamma$ | — | $0.0e+00$ |
| $\bar\sigma^{\mu\dot\alpha\alpha} = \varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\sigma^\mu_{\beta\dot\beta}$ | (38.13) | $0.0e+00$ |
| Schouten identity | (38.17) | $0.0e+00$ |

All identities verified to machine precision ($\lesssim 10^{-15}$).

# 7 Summary

| Object | Expression |
|---|---|
| $\sigma^\mu$ completeness | $\sigma^\mu_{\alpha\dot\alpha}\sigma_\mu^{\beta\dot\beta} = -2\delta_\alpha^\beta\delta_{\dot\alpha}^{\dot\beta}$ |
| Alternative form | $\varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\sigma^\mu_{\alpha\dot\alpha}\sigma^\nu_{\beta\dot\beta} = -2g^{\mu\nu}$ |
| 2-trace | $\mathrm{Tr}[\sigma^\mu\bar\sigma^\nu] = \mathrm{Tr}[\bar\sigma^\mu\sigma^\nu] = -2g^{\mu\nu}$ |
| 4-trace | $\mathrm{Tr}[\sigma^\mu\bar\sigma^\nu\sigma^\rho\bar\sigma^\sigma] = 2(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}) + 2i\varepsilon^{\mu\nu\rho\sigma}$ |
| Index lowering | $\psi_\alpha = \varepsilon_{\alpha\beta}\psi^\beta, \quad \varepsilon_{12} = -1$ |
| Index raising | $\psi^\alpha = \varepsilon^{\alpha\beta}\psi_\beta, \quad \varepsilon^{12} = +1$ |
| $\sigma \leftrightarrow \bar\sigma$ | $\bar\sigma^{\mu\dot\alpha\alpha} = \varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\sigma^\mu_{\beta\dot\beta}$ |
| Schouten | $\varepsilon_{\alpha\gamma}\varepsilon_{\beta\delta} = \varepsilon_{\alpha\beta}\varepsilon_{\gamma\delta} - \varepsilon_{\alpha\delta}\varepsilon_{\gamma\beta}$ |
| $\sigma$-Fierz | $\bar\sigma^{\mu\dot\alpha\alpha}\sigma_{\mu\beta\dot\beta} = -2\delta^\alpha_\beta\delta^{\dot\alpha}_{\dot\beta}$ |
| 4-Fermi Fierz | $(\chi^\dagger\bar\sigma^\mu\psi)(\xi^\dagger\bar\sigma_\mu\eta) = -2(\chi^\dagger\xi^\dagger)(\eta\psi)$ |

**Next:** Chapter 48 (spinors for massless particles) builds on these tools toward the spinor-helicity formalism and MHV amplitudes.